

A Radial-Twist Counterexample for Nonlinear Abel Averages

A negative answer to the open question in arXiv:1311.6698

Automated mathematics run `fa_banach_001`, lane 8

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Status. Candidate full counterexample, likely valid; human review requested.

1 Source question

Bracci, Kozitsky, and Shoikhet [1] ask the following. Let D be a balanced dense subset of the open unit ball of a complex Banach space, and let $h : D \rightarrow X$ be weakly closed and 0-dissipative. If for some $\alpha_0 > 0$ the Abel average

$$\Phi_{\alpha_0} = (\text{Id} - \alpha_0 h)^{-1}$$

exists, is holomorphic, and maps the ball into itself, must the Abel averages exist for every positive parameter?

Under the assumption that the averages converge to a holomorphic function on D , one has $\text{Null}(h) = \{(\xi, \eta) \in \mathbb{B}^2 : \lambda \xi = -\epsilon \eta^2\}$, which would imply that $\text{Null}(h)$ be a one-dimensional holomorphic retract of $\mathbb{B}^2$ (so-called *complex geodesic of $\mathbb{B}^2$*), while one-dimensional holomorphic retracts of $\mathbb{B}^2$ are known to be just the intersection of affine complex lines with $\mathbb{B}^2$.

4. OPEN QUESTION

Let D be a balanced dense subset of B , and let h be a closed (weakly closed) map on D with values in X . Assume that h is 0-dissipative and for some $\alpha_0 > 0$ its Abel average exists, is holomorphic, and maps B into itself. Do the Abel averages exist for all positive α ?

This is a full-width crop of Section 4, “Open question,” on page 13 of the source PDF.

2 Idea of the counterexample

Work in the unit disc and prescribe $\text{Id} - h$ rather than h . Inside a disc slightly larger than radius $1/2$, let $\text{Id} - h$ be the dilation $z \mapsto 2z$. Outside, keep its output modulus equal to $2|z|$ but gradually rotate it through an angle of at most $\pi/3$. The modulus $2|z|$ makes $\text{Id} - h$ globally injective, while the absence of rotation on the preimage of the unit disc makes its Abel average exactly $w \mapsto w/2$.

The rotation is small enough to preserve dissipativity: its cosine is at least $1/2$. But at parameter $1/2$, the map is the midpoint of the identity and the twisted dilation. Two carefully chosen circles

then have the same image radius, and their angular phases can be aligned. This creates an exact two-point collision over a point strictly inside the unit disc.

3 Construction and theorem

Let $X = \mathbb{C}$ with its usual norm, let $\mathbb{D} = \{z : |z| < 1\}$, and take $D = \mathbb{D}$. Set

$$a = \frac{\sqrt{7}}{5}, \quad b = \frac{3}{5}.$$

Thus $1/2 < a < b < 1$. Define a continuous function $\varphi : [0, 1) \rightarrow [0, \pi/3]$ by

$$\varphi(r) = \begin{cases} 0, & 0 \leq r \leq a, \\ \frac{\pi}{3} \frac{r - a}{b - a}, & a < r < b, \\ \frac{\pi}{3}, & b \leq r < 1. \end{cases}$$

Finally define

$$h(z) = (1 - 2e^{i\varphi(|z|)})z, \quad z \in \mathbb{D}.$$

Theorem 1. *The map $h : D \rightarrow \mathbb{C}$ is weakly closed and 0-dissipative. Its Abel average at $\alpha_0 = 1$ exists and is the holomorphic self-map*

$$\Phi_1(w) = \frac{w}{2}, \quad w \in \mathbb{D}.$$

Nevertheless, the Abel average at $\alpha = 1/2$ does not exist: the map $\text{Id} - \frac{1}{2}h$ is not injective over $\mathbb{D}$. Consequently the open question of [1] has a negative answer.

Proof. The domain $D = \mathbb{D}$ is balanced and dense in $\mathbb{D}$. Since φ is continuous, so is h . In finite dimension weak and norm convergence agree, so h is weakly closed (indeed, it has a norm-closed graph).

Identify $\mathbb{C}^*$ with $\mathbb{C}$ in the usual way. For $z \neq 0$, the unique element of $J(z)$ is the functional $z^*(w) = \bar{z}w$. Therefore

$$\text{Re}\langle h(z), z^* \rangle = (1 - 2\cos\varphi(|z|))|z|^2 \leq 0,$$

because $0 \leq \varphi \leq \pi/3$. Hence the definition in [1] holds with $\varsigma = 0$ (in fact the inequality holds throughout the disc), and h is 0-dissipative.

At parameter 1 we have

$$(\text{Id} - h)(z) = 2e^{i\varphi(|z|)}z. \tag{1}$$

This map is globally injective on $\mathbb{D}$: equality of two outputs first gives equality of their moduli, since the output modulus is $2|z|$; the two values of φ then agree, and (1) gives equality of the inputs. In fact (1) maps $\mathbb{D}$ bijectively onto the disc of radius 2: for a point w in that disc, put $r = |w|/2$ and $z = \frac{1}{2}e^{-i\varphi(r)}w$.

For $w \in \mathbb{D}$, its unique preimage under (1) has modulus $|w|/2 < 1/2 < a$. On that region $\varphi = 0$, so the preimage is simply $w/2$. Thus

$$\Phi_1 = (\text{Id} - h)^{-1}|_{\mathbb{D}} : w \mapsto \frac{w}{2}$$

is holomorphic and maps $\mathbb{D}$ into itself, exactly as required by the source hypothesis.

It remains to show failure at parameter $1/2$. We have

$$\left(\text{Id} - \frac{1}{2}h\right)(z) = \left(\frac{1}{2} + e^{i\varphi(|z|)}\right)z. \quad (2)$$

Choose $\psi \in (0, \pi/2)$ by

$$e^{i\psi} = \frac{2 + i\sqrt{3}}{\sqrt{7}},$$

and set

$$z_1 = \frac{\sqrt{7}}{5}, \quad z_2 = \frac{3}{5}e^{-i\psi} = \frac{3(2 - i\sqrt{3})}{5\sqrt{7}}.$$

Their moduli are a and b , respectively, so $\varphi(|z_1|) = 0$ and $\varphi(|z_2|) = \pi/3$. Formula (2) gives

$$\left(\text{Id} - \frac{1}{2}h\right)(z_1) = \frac{3}{2} \frac{\sqrt{7}}{5} = \frac{3\sqrt{7}}{10}.$$

Also

$$\frac{1}{2} + e^{i\pi/3} = \frac{2 + i\sqrt{3}}{2} = \frac{\sqrt{7}}{2}e^{i\psi},$$

and hence

$$\left(\text{Id} - \frac{1}{2}h\right)(z_2) = \frac{\sqrt{7}}{2}e^{i\psi} \frac{3}{5}e^{-i\psi} = \frac{3\sqrt{7}}{10}.$$

The points z_1 and z_2 are distinct because their moduli differ, whereas their common image belongs to $\mathbb{D}$ since

$$\left|\frac{3\sqrt{7}}{10}\right|^2 = \frac{63}{100} < 1.$$

Thus $(\text{Id} - \frac{1}{2}h)^{-1}$ is not a function on $\mathbb{D}$, so the Abel average at $\alpha = 1/2$ does not exist. This proves the negative answer. $\square$

4 Checks, scope, and relation to the partial theorem

The construction satisfies a stronger property than the source hypothesis: $\text{Id} - h$ is globally injective on D and even maps D bijectively onto the disc of radius 2. Thus the counterexample is insensitive to whether the inverse notation is read globally or only over the unit ball.

The earlier upward-propagation result for this target remains consistent with the example: existence at $\alpha_0 = 1$ forces all averages for $\alpha \geq 1$, while the present failure occurs at $\alpha = 1/2$. The example therefore shows that the unresolved downward direction is a genuine obstruction and settles the full source question negatively.

The optional script `code/check_counterexample.py` samples the dissipativity inequality and evaluates the collision numerically. It is not part of the proof; all calculations above are exact.

5 Novelty check and review recommendation

A bounded search covered the exact printed question, arXiv id 1311.6698, the source title and authors, and combinations of “weakly closed,” “0-dissipative,” “Abel average,” “counterexample,” and “negative answer.” No later answer or matching construction was found. The 2019 monograph

[2] reproduces the question as open. Novelty confidence is therefore moderate-high within the searched sources, but exhaustive priority is not claimed.

Human review should focus on the support-functional calculation, global injectivity of (1), and the two exact evaluations of (2). No unproved lemma or external theorem is used.

References

- [1] F. Bracci, Y. Kozitsky, and D. Shoikhet, *Abel averages and holomorphically pseudo-contractive maps in Banach spaces*, J. Math. Anal. Appl. 423 (2015), no. 2, 1580–1593; arXiv:1311.6698. DOI: 10.1016/j.jmaa.2014.10.079.
- [2] M. Elin, S. Reich, and D. Shoikhet, *Numerical Range of Holomorphic Mappings and Applications*, Birkhäuser, Cham, 2019. DOI: 10.1007/978-3-030-05020-7.